

Shamar Weekes

shamar-weekes.com | weekesshamar38@gmail.com | linkedin.com/in/shamar-weekes | github.com/shamar12334

EDUCATION

Florida International University

Bachelor of Science in Computer Science

Expected Graduation: December 2026

Miami, FL

- **Relevant Coursework:** Programming II, Data Structures, Systems Programming, Principles of Programming Languages, Software Engineering I, Computer Architecture, Technology in the Global Arena
- **Organizations:** CodeCrunch, INIT, BreakThrough Tech, Panther Robotics, Women in Computer Science (WiCS)

EXPERIENCE

Founder & Software Engineer – Patient Care Validation System

November 2025 – Present

Independent Project | Family Healthcare Network (4 Locations: FL, TX, IL, MA)

- **Design and deploy a BLE-based indoor tracking system across 4 healthcare facilities (120+ beds, 20+ staff)** to automatically log staff-patient interactions and validate care delivery.
- **Build a digital floor-mapping system** to model rooms, beds, and care zones, enabling real-time indoor positioning and event tracking.
- **Engineer a real-time event pipeline** capturing visit timestamps, duration, and frequency, eliminating manual documentation and improving audit readiness.
- **Develop compliance monitoring dashboards** to identify missed or delayed patient checks, increasing operational visibility for administrators.
- **Architect scalable backend services (FastAPI + PostgreSQL)** to process high-frequency BLE signals and support real-time querying.

Open Source Contributor

August 2025 – Present

Python, React, TensorFlow, Kestra

- **Merged PR #7321** into Kestra (10k+ GitHub stars), resolving a legend component bug affecting UI rendering across the platform.
- Iterated across 3 pull requests to diagnose and isolate the root cause, demonstrating persistence and attention to detail in a large production codebase.
- Collaborate with maintainers and engineers through code reviews, incorporating feedback and iterating on solutions in real-world development workflows.
- Exploring the TensorFlow codebase to identify opportunities for bug fixes and feature contributions, gaining hands-on familiarity with a production ML framework.

PROJECTS

Python, TensorFlow,

Real-Time Human Activity Recognition | Personal Project | March 2026 – Present

OpenCV, Flask

- Built a real-time human activity recognition system using TensorFlow and OpenCV to classify actions like walking, running, sitting, and jumping with 92% accuracy on personal webcam data.
- Collected and annotated 1,200+ video clips from personal workouts and daily activities to train and validate the model.
- Designed a Python + Flask dashboard displaying live predictions, confidence scores, and activity duration in real-time.
- Optimized model with dropout, early stopping, and data augmentation, reducing misclassification by 28% and training time by 35%.

C/C++, Embedded Systems, Sensors, Microcontrollers

FIU-SEDS | Vensors | January 2025 – April 2025

- Develop and integrate code for the Vensors project, controlling flight sensors and accelerometers on a microcontroller platform.
- Adjust and optimize full system setup through PRs (16, 25), reducing initialization errors by **90%** and ensuring accurate data collection across **3+ microcontrollers**.
- Implement flight sensor and acceleration calculation functions, achieving **100% measurement accuracy** on prototype test models over **50+ test cycles**.
- Automate calibration and validation process, reducing setup and testing time from **2 hours to 15 minutes per session**.
- Collaborate with a team of **9+ engineers**, contributing to debugging, code integration, and iterative system improvements, accelerating overall development timeline by **30%**.

LEADERSHIP

Workshop Lead, CodeCrunch

September 2025 – December 2025

Miami, FL / Zoom

- Led twice-weekly hybrid networking workshops for FIU students, with hands on practice and lecture slides.
- Developed all curriculum materials and exercises from scratch, to facilitate student learning and engagement.

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++, C# , SQL, HTML, CSS

Frameworks & Libraries: React, Next.js, Node.js, Express, Django, Flask, FastAPI, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Microsoft SQL Server, Redis, Firebase

Tools & Technologies: Docker, Git, CI/CD, Linux, REST APIs, PyTest, JUnit, Vitest, Figma, Agile, OOP